An FPGA-Accelerated AI-Assisted Precision Remote-Controlled Perimeter Security System (PRC-PSS) for Enhanced Border Security

Platform 2

Abstract

National border security remains a demanding challenge where soldiers are often placed in harm's way. To address this, we propose the development of an FPGA-accelerated Precision Remote-Controlled Perimeter Security System (PRC-PSS), a prototype centered on a custom System-on-Chip (SoC) design implemented on the Microchip PolarFire SoC Icicle Kit. Our core innovation is the design of a low-power Convolutional Neural Network (CNN) hardware accelerator on the FPGA fabric to assist our soldier in automating the tracking process of the gun for a clear headshot of the enemy. This accelerator will execute YOLO-style object detection and DisNet-style distance estimation with real-time, minimal latency that is unachievable in a purely software-based system. This AI-driven capability automates the critical task of threat identification, providing the soldier with instant situational awareness to make faster and safer decisions. By partitioning the system, the RISC-V CPU subsystem will manage high-level tasks while our custom accelerator performs the computationally intensive AI inference. In this prototype, a laser pointer serves as a non-lethal deterrent, allowing us to validate the algorithms for hardware-accelerated trajectory approximation and precise servo motor control. A soldier's confirmation remains mandatory, ensuring a human-in-the-loop control mechanism. The successful development of this prototype will validate a power-efficient, highly accurate hardware architecture that can save soldier lives by reducing the need for direct exposure.

Main Idea

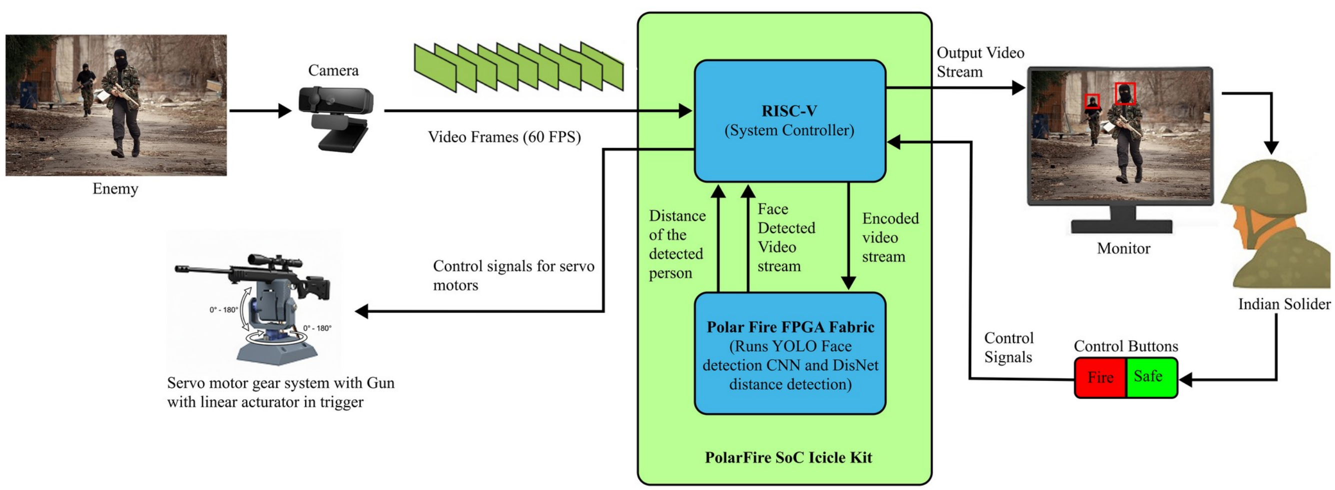

Fig. 1 System architecture of the FPGA-accelerated RISC-V based PRC-PSS, showing video acquisition, YOLO CNN-based face detection, distance calculation by DisNet, annotated video streaming, and human-in-the-loop control.

The PRC-PSS operates on a hardware/software co-design principle as shown in Fig. 1, partitioning tasks between the RISC-V CPU and custom FPGA logic to maximize efficiency and real-time performance. Initially, the real-time video stream from the camera is managed by the RISC-V CPU subsystem, which handles the input pipeline and prepares the frames for analysis. To identify and track hostile targets, the CPU offloads the computationally intensive inference task to our custom-designed CNN hardware accelerator [1] implemented on the PolarFire SoC's FPGA fabric. This accelerator is architected to efficiently execute the core layers of both the YOLO object detection [2] and DisNet [3] distance estimation algorithms, processing each frame to generate bounding box coordinates and depth data with minimal latency. The resulting positional and distance data is then returned to the RISC-V CPU, which performs the final trajectory calculations. This high-speed loop repeats for every frame, enabling smooth and continuous tracking of the enemy as it moves.

The system's aiming mechanism is driven by servo motors, which are electromechanical devices that can rotate up to 180 degree angle and provide precise rotational control in response to electrical signals. Based on the target's coordinates, the module sends these signals to the servo motors, adjusting their angles to align the gun barrel with the target. To achieve even finer control over aiming, a high-precision gear system is coupled to the servo motors [4,5]. By meshing these gears, the system translates the motor's rotation into smaller, more precise angular movements [6,7], significantly improving aiming accuracy for the minute adjustments required for a successful shot. Finally, once the aiming process is complete, the module commands the linear actuator to fire the weapon, after receiving a soldier's approval, the system fires the weapon at the confirmed target [8]. For the prototype model, a laser pointer is used to indicate the target instead of an actual gun. This allows for safe testing and validation of the aiming algorithm's precision.

APPLICATION

- Border checkpoint monitoring to detect unauthorized incursions.
- Autonomous military vehicles for reconnaissance and combat missions.
- Securing military bases to provide enhanced surveillance and threat assessment.
- Protecting critical infrastructure, such as power plants or government buildings

VALUE ADD

- Enhanced Soldier Safety: By physically separating the operator from the line of fire, the system dramatically reduces the risk of soldier casualties and the psychological strain of combat.
- Superior Accuracy and Efficiency: The combination of YOLO and DisNet, coupled with a precise motor and gear system, provides an unmatched level of aiming accuracy. The AI's ability to autonomously identify and track targets frees up the operator to focus on strategic decision-making and battlefield management.
- Ethical and Legal Framework: The system is designed with a mandatory human-in-the-loop confirmation step, directly addressing ethical concerns about autonomous weapons and ensuring accountability and human oversight in the use of lethal force.
- For future production, the entire FPGA logic can be implemented as a single custom chip (ASIC), creating an even more compact and robust final system.

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